

Separating Network and Inverter-Control Effects on Damping Margin via a Rational Graph-Filter Perturbation

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Abstract—The replacement of synchronous generation by inverter-based resources changes not only the inertia of a power grid but also the geometry of its damping. Grid-following and grid-forming controls introduce non-proportional damping and additional controller states, so a graph-frequency improvement can still reduce damping margin. This paper develops a second-order spectral perturbation framework that separates the network contribution to damping margin from the inverter-control contribution. The zero-order model places inertial-Laplacian modes in the damping-region plane (ν_k, δ_k) . The central result is a closed-form second-order correction for the critical pole under off-diagonal modal damping,

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell} E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell},$$

where E is the off-diagonal part of $Q^T \tilde{D} Q$. This correction is also a rational graph filter of the operator pair $(\tilde{L}, \tilde{D})$ evaluated at the critical pole. The filter form explains why polynomial graph filters of $\tilde{L}$ alone cannot represent IBR damping: the missing information is the modal coupling created by non-proportional damping. Synthetic tests show that the correction substantially improves the diagonal screen in weak-to-moderate non-proportional regimes, while clustered or strongly coupled regimes still require a reduced QEP. A reduced seven-state grid-following inverter model with PLL memory confirms the physical interpretation: treating the PLL as memory-free gives a median damping-margin error of 2.16%, while the second-order correction reduces it to 0.09% over 399 stable cases. An IEEE 39-bus ANDES baseline, an IEEE39-derived Kron-reduced IBR surrogate, and first SG-to-GFL/GFM replacement pilots are reported as benchmark-construction steps. The IBR pilots are not yet calibrated estimator validation, so full IBR benchmark claims remain pending.

Index Terms—Damping ratio, inverter-based resources, eigenvalue perturbation, phase-locked loop, quadratic eigenvalue problem, rational graph filter, graph signal processing, small-signal stability.

I. INTRODUCTION

Power systems are moving from synchronous-machine-dominated dynamics toward networks with high penetration of inverter-based resources (IBRs). The most visible effect of this transition is reduced inertia, but the more difficult small-signal effect is that damping is no longer proportional to inertia

or to network stiffness. A synchronous generator contributes inertia and damping through electromechanical dynamics. A grid-following inverter contributes control damping through a phase-locked loop and current controller; a grid-forming inverter contributes droop or virtual-synchronous-machine behavior. These mechanisms are physically different and appear in the linearized model as heterogeneous, non-commuting dynamic coefficients.

The standard swing-equation intuition says that low-frequency inter-area modes are critical and that increasing network stiffness improves robustness. Those statements are useful but incomplete. The damping ratio of a mode is not simply a graph frequency. If a line reinforcement raises the oscillation frequency without increasing the modal decay rate enough, the damping ratio can decrease. Likewise, if damping is heterogeneous, the mode with lowest damping ratio need not be the Fiedler mode or the highest-frequency mode. Planning rules based only on λ_2 , Fiedler participation, or frequency sensitivities can therefore be misleading in an IBR-rich grid.

This paper asks a narrow question: how much of the damping margin is explained by the electromechanical network, and how much is moved by inverter-control coupling? The answer proposed here is a perturbation-based damping-margin estimator. It starts from the inertial graph modes of the linearized network, treats the diagonal modal damping prediction as the zero-order margin, and then adds a closed-form second-order correction caused by non-proportional modal coupling. A reduced quadratic eigenvalue problem (QEP) is retained as the fallback when the perturbation is unsafe.

The central message is not that a new graph eigenvalue replaces small-signal analysis. The central message is that a calibrated spectral perturbation can expose which part of small-signal stability is carried by the network and which part is changed by inverter damping and control. This distinction matters for planning: the right bus to convert from synchronous generation to IBR is not generally the bus with the smallest Fiedler participation, and the right line to reinforce is not necessarily the line with largest $\partial \nu_k / \partial w_{ij}$.

The paper makes four calibrated contributions. The novelty claim is at the level of power-system formulation, interpretation, and validated result, not at the level of inventing eigenvalue perturbation theory.

- It uses the damping-region plane (ν_k, δ_k) as the zero-order network picture. This is an interpretive coordinate system for heterogeneous IBR damping, not a claim that the proportional modal formula is new.

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This manuscript is a pre-submission research draft. The method and synthetic stress tests are reported explicitly; full IEEE 39-bus validation with calibrated ANDES IBR models remains a required validation stage before any final journal submission.

- It derives and tests a closed-form second-order correction for the critical damping-margin pole under off-diagonal modal damping. This correction explains, term by term, how non-proportional inverter damping couples the critical mode to other modes.
- It organizes the estimator as a diagnostic workflow: diagonal screen, second-order correction, and reduced-QEP fallback for clustered or strongly coupled regimes. The negative cases where second order worsens the tail are part of the method, because they justify the trigger.
- It shows how the same margin quantity can support planning applications, including SG-to-IBR conversion, virtual damping/inertia placement, and damping-aware line reinforcement. These are applications of the core estimator and must be compared against existing damping-ratio placement and QEP-sensitivity literature.

The present manuscript is deliberately calibrated. Proportional modal damping, Rayleigh sensitivity, QEP sensitivity, and eigenvalue perturbation are established tools. The candidate contribution is to use second-order perturbation to decompose damping margin into network and inverter-control effects, then validate when that cheap correction is reliable and when QEP fallback is needed. Synthetic stress tests and a reduced PLL model are reported to identify mechanisms and failure modes. The IEEE 39-bus ANDES protocol is specified because no final Transactions submission should claim benchmark validity without it.

II. RELATED WORK

Small-signal power-system stability is a mature field; standard references include modal analysis of electromechanical oscillations and damping-ratio sensitivity [1], [2]. Spectral and system-theoretic approaches to power networks have clarified the role of Laplacian eigenvalues, inertia, and damping in low-inertia performance [3]–[5]. In the proportional or homogeneous case, modal frequencies scale as $\sqrt{\nu_k}$ and damping ratios take a closed form involving damping, inertia, and graph eigenvalues. Those results are essential background for this paper rather than its novelty.

IBR-dominated systems complicate this picture because their controls introduce additional states, time scales, and non-proportional damping. Recent small-signal studies of inverter networks emphasize the role of line dynamics, operating conditions, and converter-control parameters [6]–[8]. Stability-region and security-region viewpoints for converter-dominated systems are also active research topics [9], [10]. This activity creates both an opportunity and a risk: region language is useful, but broad claims of originality are unsafe unless the exact reduced spectral construction is differentiated carefully.

The proposed damping-region picture is inspired by consensus-region theory in multi-agent systems. In that literature, stability of a networked linear system can be tested by asking whether scaled nonzero Laplacian eigenvalues lie in a region determined by local agent dynamics and feedback gains [11]. Related networked-control books develop dynamic topologies, switching graphs, dwell-time logic, resilient consensus, delays, and event-triggered control [12], [13]. The

present paper does not import those results wholesale. A power grid with IBR controls is not a first-order consensus system. The transfer is conceptual: define the region induced by local dynamics and use graph-modal coordinates to screen network-level stability.

Quadratic eigenvalue problems provide the natural algebraic language for second-order small-signal models [14]. If damping and stiffness commute, the QEP decouples into scalar modal equations. If not, modes are coupled through off-diagonal modal damping terms. Modal perturbation for non-proportionally damped systems is well established in structural dynamics [15], [16]; this paper does not claim that perturbation theory itself is new. The proposed distinction is the power-system use of the second-order term as a damping-margin decomposition and as a rational graph filter of $(\tilde{L}, \tilde{D})$.

Planning-oriented damping metrics also have nearby prior art. Grid-forming inverter placement by composite frequency and damping indices has been studied on modified IEEE 39-bus systems [17]. Closed-form damping-ratio sensitivities based on QEP formulations have also been derived for power-system remedial actions [18]. The line-reinforcement and weak-node maps in this manuscript must therefore be read as applications of the proposed margin object and must be compared against these baselines. The validation platform used here is ANDES, an open-source symbolic-numeric power-system modeling tool with power flow, time-domain, and eigenvalue analysis [19], [20].

III. NETWORK MODEL AND DAMPING REGION

Consider a lossless small-signal power network linearized around an operating point:

$$M\ddot{\theta} + D\dot{\theta} + L\theta = p, \quad (1)$$

where $M = \text{diag}(M_i) \succ 0$, $D = \text{diag}(D_i) \succeq 0$, and L is the weighted Laplacian induced by the active-power Jacobian. In the usual approximation,

$$L_{ij} = -V_i V_j b_{ij} \cos(\theta_i^* - \theta_j^*), \quad i \neq j, \quad (2)$$

with diagonal entries chosen so that $L\mathbf{1} = 0$. The analysis is performed in inertia-normalized coordinates $\vartheta = M^{1/2}\theta$, giving

$$\ddot{\vartheta} + \tilde{D}\dot{\vartheta} + \tilde{L}\vartheta = M^{-1/2}p, \quad (3)$$

where

$$\tilde{L} = M^{-1/2}LM^{-1/2}, \quad \tilde{D} = M^{-1/2}DM^{-1/2}. \quad (4)$$

Let $\tilde{L}q_k = \nu_k q_k$, with $Q = [q_1, \dots, q_n]$ orthonormal. The zero eigenvalue associated with uniform angle shift is discarded. In modal coordinates $x = Q^T \vartheta$, the dynamics are

$$\ddot{x} + \Gamma\dot{x} + \Lambda x = Q^T M^{-1/2}p, \quad (5)$$

where $\Gamma = Q^T \tilde{D}Q$ and $\Lambda = \text{diag}(\nu_k)$. The diagonal entries

$$\delta_k = \Gamma_{kk} = q_k^T \tilde{D}q_k \quad (6)$$

are modal damping coefficients, while the off-diagonal entries $\Gamma_{\ell k}$ describe damping-induced modal coupling.

TABLE I
NEAREST PRIOR ART AND NOVELTY BOUNDARY. THE PAPER DOES NOT CLAIM NOVELTY FOR GENERIC PERTURBATION THEORY, QEP SENSITIVITY, OR DAMPING-RATIO PLACEMENT METRICS.

Reference area	What is already established	What is not claimed here	Candidate distinction here
Non-proportional modal perturbation [15], [16]	Perturbation methods and indices for non-classically damped second-order systems.	The perturbation expansion itself is not new.	Use the second-order term as a damping-margin decomposition for IBR grids and express it as a rational filter of $(\tilde{L}, \tilde{D})$.
GFM placement by stability indices [17]	Placement/ranking of grid-forming inverters on modified IEEE 39 cases using frequency and damping-related indices.	Ranking inverter placement by a damping metric is not new.	Compare conversion decisions using the proposed margin estimator and report regret against full eigensolve baselines.
QEP damping-ratio sensitivities [18]	Closed-form QEP sensitivities for damping-ratio remedial actions in synchronous-generator models.	QEP sensitivity formulas are not new.	Apply damping-ratio sensitivity to line reinforcement and IBR-oriented margin maps, with frequency-only ranking as a negative control.

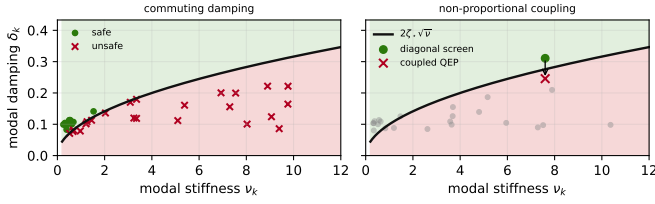


Fig. 1. Damping-region view. Left: in the commuting case, a target ζ_* becomes the exact boundary $\delta = 2\zeta_*\sqrt{\nu}$. Right: in non-proportional damping, a diagonal modal point can appear safe while QEP coupling moves the pole below the margin. The region is therefore a screen, not a certificate, outside the commuting regime.

The exact small-signal poles are roots of the QEP

$$P(s)v = (s^2I + s\tilde{D} + \tilde{L})v = 0. \quad (7)$$

Equivalently, they are eigenvalues of the first-order matrix

$$A = \begin{bmatrix} 0 & I \\ -M^{-1}\tilde{L} & -M^{-1}\tilde{D} \end{bmatrix}. \quad (8)$$

The damping margin is

$$\zeta_{\min} = \min_{s_j \in \mathcal{S}_{\text{osc}}} \frac{-\text{Re}(s_j)}{|s_j|}, \quad (9)$$

where $\mathcal{S}_{\text{osc}}$ contains complex oscillatory poles.

A. Damping Region

If $\tilde{L}$ and $\tilde{D}$ commute, then Γ is diagonal and each nonzero mode solves

$$s^2 + \delta_k s + \nu_k = 0. \quad (10)$$

For an underdamped mode,

$$\zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (11)$$

Thus a target damping margin ζ_* is equivalent to

$$\delta_k \geq 2\zeta_*\sqrt{\nu_k}, \quad k = 2, \dots, n. \quad (12)$$

Equation (12) defines the damping region shown in Fig. 1. Modes below the curve violate the target damping margin. Modes above it satisfy the target in the proportional case.

This region should not be overinterpreted. In the non-proportional case, Γ is not diagonal, and (11) is no longer exact. It remains a first-order approximation when damping

is weak and modal frequencies are well separated. It can fail when low-damping modes are clustered or when $\Gamma_{\ell k}$ is large relative to the modal frequency separation. This observation motivates an adaptive estimator rather than a single formula.

Proposition 1 (Exact region in the commuting case): Suppose $\tilde{L}$ and $\tilde{D}$ are symmetric and share an orthonormal eigenbasis on the nonzero-angle subspace. Then the damping margin of (3) is

$$\zeta_{\min} = \min_{\nu_k > 0} \frac{\delta_k}{2\sqrt{\nu_k}}, \quad (13)$$

provided the limiting modes are underdamped. Moreover, a target margin ζ_* is satisfied if and only if every nonzero modal point obeys

$$\delta_k \geq 2\zeta_*\sqrt{\nu_k}. \quad (14)$$

The proof follows by diagonalizing both matrices in (3). It is stated explicitly because it fixes the boundary of the proposed framework. The region is exact only when damping is diagonal in the graph-modal basis. In every other case, the same inequality is a screen, not a certificate.

Proposition 2 (First-order modal-coupling warning): Let ν_k and ν_ℓ be two distinct inertial-Laplacian eigenvalues and let $\Gamma_{\ell k} \neq 0$. The sensitivity of modal damping to a perturbation that rotates q_k toward q_ℓ scales with

$$\frac{\Gamma_{\ell k}}{\nu_k - \nu_\ell}. \quad (15)$$

Thus clustered graph frequencies and large off-diagonal modal damping are the cases in which the diagonal region is least reliable.

This warning is diagnostic, not a replacement for the full QEP. It explains why two networks with the same diagonal modal points can have different damping margins: the missing information is the coupling geometry in $Q^T \tilde{D} Q$, weighted by modal separation. This is the mechanism behind the adaptive estimator in the next section.

Remark 1 (Critical mode selection): For damping ratio, the critical graph mode is not generically the Fiedler mode. In proportional damping with homogeneous damping per inertia, the smallest damping ratio may be controlled by the largest ν_k because ζ_k scales like $1/\sqrt{\nu_k}$. In heterogeneous IBR cases, the critical mode is determined by the pair (ν_k, δ_k) and by modal coupling. Any benchmark that compares only the Fiedler-mode estimate against the full eigensolve is therefore using the wrong control.

IV. ADAPTIVE SPECTRAL MARGIN ESTIMATOR

The estimator uses the diagonal region as a cheap screen, a second-order perturbation as the closed-form correction, and the reduced QEP as a fallback. The diagonal all-mode estimate is

$$\hat{\zeta}_{\text{diag}} = \min_{k:\nu_k>0} \frac{q_k^T \tilde{D} q_k}{2\sqrt{\nu_k}}. \quad (16)$$

This is not the Fiedler-mode estimate. It takes the minimum over all modes. In homogeneous damping the critical mode is often the largest-frequency mode; in heterogeneous damping it need not be.

A. Closed-Form Second-Order Correction

Let $\Gamma = \Delta + E$, where $\Delta = \text{diag}(\delta_k)$ and E contains the off-diagonal modal damping. The zero-order diagonal pole of the critical mode c is s_0 , where

$$s_0^2 + \delta_c s_0 + \nu_c = 0. \quad (17)$$

Let $p_\ell(s) = s^2 + \delta_\ell s + \nu_\ell$. Partition the modal coordinates into the critical coordinate z_c and all remaining coordinates z_R . To second order in the off-diagonal damping, the coupled modal equations are

$$p_c(s)z_c + sE_{cR}z_R = 0, \quad (18)$$

$$\text{diag}(p_\ell(s))z_R + sE_{Rc}z_c = 0, \quad (19)$$

where the E_{RR} block would contribute only higher-order terms to the scalar critical equation. Thus

$$z_R = -s \text{diag}(p_\ell(s))^{-1} E_{Rc} z_c + O(\|E\|^2). \quad (20)$$

Substitution into the critical equation gives

$$p_c(s) - s^2 E_{cR} \text{diag}(p_\ell(s))^{-1} E_{Rc} = 0 + O(\|E\|^3). \quad (21)$$

Writing $s = s_0 + \Delta s_c$ and evaluating the resolvent at the critical pole s_0 yields the second-order pole correction

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{E_{c\ell} E_{\ell c}}{s_0^2 + \delta_\ell s_0 + \nu_\ell}. \quad (22)$$

The corrected pole is $s_c^{(2)} = s_0 + \Delta s_c$ and the corrected margin is

$$\hat{\zeta}_2 = -\frac{\text{Re}(s_c^{(2)})}{|s_c^{(2)}|}. \quad (23)$$

Equation (22) is the central estimator in this manuscript. It is not a new perturbation theorem; it is a power-system use of second-order eigenvalue perturbation to quantify how non-proportional inverter damping moves the critical damping-margin pole away from the diagonal network prediction.

B. Joint Inertia–Damping Perturbation

The previous correction assumes that $\tilde{L}$ is fixed and that only off-diagonal modal damping perturbs the diagonal QEP. A finite SG-to-IBR conversion, however, changes both inertia and damping. Let a small conversion parameter ρ define

$$M(\rho) = M_0 + \rho \Delta M, \quad D(\rho) = D_0 + \rho \Delta D, \quad (24)$$

and define

$$\Lambda_1 = Q^T \frac{d\tilde{L}}{d\rho} Q, \quad \Gamma_1 = Q^T \frac{d\tilde{D}}{d\rho} Q \quad (25)$$

in the unperturbed inertial-Laplacian basis. The first-order shift of the critical diagonal pole is

$$s_1 = -\frac{\Lambda_{1,cc} + s_0 \Gamma_{1,cc}}{2s_0 + \delta_c}. \quad (26)$$

The local pole approximation has the form

$$s_c = s_0 + s_1 + s_{2,\text{coup}} + \text{diagonal self terms} + O(\rho^3). \quad (27)$$

The next equation isolates only the modal-coupling part of the second-order term; it is not the complete finite-conversion pole shift. The off-diagonal second-order coupling contribution is obtained by replacing the perturbation sE in (22) with the combined channel

$$H_1(s_0) = s_0 \Gamma_1 + \Lambda_1. \quad (28)$$

Thus

$$s_{2,\text{coup}} = \frac{1}{2s_0 + \delta_c} \sum_{\ell \neq c} \frac{N_{c\ell}(s_0)}{s_0^2 + \delta_\ell s_0 + \nu_\ell}, \quad (29)$$

$$N_{c\ell}(s_0) = \left(s_0 E_{c\ell}^{(D)} + E_{c\ell}^{(L)} \right) \left(s_0 E_{\ell c}^{(D)} + E_{\ell c}^{(L)} \right). \quad (30)$$

where $E^{(D)} = \text{offdiag}(\Gamma_1)$ and $E^{(L)} = \text{offdiag}(\Lambda_1)$. If $\Delta M = 0$, then $E^{(L)} = 0$ and (29) collapses to (22). Expanding the numerator exposes the non-additive term

$$s_0 \left(E_{c\ell}^{(D)} E_{\ell c}^{(L)} + E_{c\ell}^{(L)} E_{\ell c}^{(D)} \right), \quad (31)$$

which is the explicit red-control interaction: damping and inertia perturbations do not generally move the critical pole as two separable scalar effects.

Equation (29) is used here as a mechanism statement and as a local score for small virtual-inertia or droop retuning. It should not be presented as a reliable estimator for a large finite SG-to-IBR conversion. For such conversions, the adaptive estimator or the full model must be recomputed after each candidate action. The full second-order expansion also contains scalar self terms from the diagonal first-order shift s_1 ; (29) isolates the modal-coupling contribution that generalizes the rational-filter correction.

C. Rational Graph-Filter Form

The same correction can be written in graph-signal-processing language. A polynomial filter of the inertial Laplacian,

$$h(\tilde{L}) = \sum_{m=0}^{M_h} a_m \tilde{L}^m, \quad (32)$$

acts diagonally in the eigenbasis of $\tilde{L}$: $h(\tilde{L})q_k = h(\nu_k)q_k$. It can rescale modal components, but it cannot create the off-diagonal modal damping entries that are absent from a graph-frequency-only description. This is the structural reason why a polynomial filter of $\tilde{L}$ alone is not a damping-margin model for a non-proportionally damped IBR grid.

Define the pole-dependent rational filter

$$H(\Lambda; s_0) = \text{diag} \left(\frac{1}{s_0^2 + \delta_\ell s_0 + \nu_\ell} \right)_{\ell \neq c}, \quad (33)$$

with the critical index omitted from the diagonal action. Then (22) is equivalently

$$\Delta s_c = \frac{s_0^2}{2s_0 + \delta_c} [EH(\Lambda; s_0)E]_{cc}. \quad (34)$$

Proposition 3 (Rational pair-filter interpretation): The second-order margin correction is a rational filter of the dynamic operator pair $(\tilde{L}, \tilde{D})$ evaluated at the diagonal critical pole. It depends on $\tilde{L}$ through the modal frequencies ν_ℓ , on $\tilde{D}$ through δ_ℓ and $E = \text{offdiag}(Q^T \tilde{D} Q)$, and on the dynamics through s_0 . It is therefore not reducible to a polynomial filter of $\tilde{L}$.

Proof: Equation (22) can be factored by collecting the modal denominators into the diagonal operator $H(\Lambda; s_0)$. The left and right multiplications by E inject and recover the non-proportional damping coupling. If $E = 0$, the filter output is zero and the proportional damping-region formula is exact. If $E \neq 0$, the correction weights each coupled mode by its resolvent-like separation from the critical pole. $\square$

This filter form is an interpretation of (22), not a second independent mechanism. Its value is that it states precisely what graph information is missing from $\tilde{L}$ alone: damping margin is governed by the pair $(\tilde{L}, \tilde{D})$ and by the pole at which the pair is evaluated.

The correction is used only when the perturbation is credible. If $|s_0^2 + \delta_\ell s_0 + \nu_\ell|$ is small for a neighboring mode, or if $|\Delta s_c|$ is comparable to $|s_0|$, the method should reject the correction and solve a reduced QEP instead.

Define a simple coupling score for a candidate mode set $\mathcal{I}$:

$$\eta(\mathcal{I}) = \max_{\substack{k, \ell \in \mathcal{I} \\ k \neq \ell}} \frac{|\Gamma_{\ell k}|}{|\sqrt{\nu_\ell} - \sqrt{\nu_k}| + \epsilon}, \quad (35)$$

where $\epsilon > 0$ prevents division by zero. Large η indicates that modal damping couples frequencies too strongly for a purely diagonal approximation. More elaborate diagnostics can include the coupling to nearby modes outside $\mathcal{I}$ and the residual of the reduced QEP.

Let $\mathcal{I}_r$ contain the modes with the smallest diagonal damping estimates and any neighboring modes triggered by (35). With $V = Q[:, \mathcal{I}_r]$, form

$$L_r = V^T \tilde{L} V, \quad D_r = V^T \tilde{D} V, \quad (36)$$

and solve

$$(s^2 I_r + s D_r + L_r) w = 0. \quad (37)$$

The adaptive estimate returns the diagonal estimate when the coupling diagnostic is small and the reduced-QEP estimate otherwise.

D. Synthetic Stress Test

The purpose of the first numerical experiment is not to claim benchmark validity, but to identify mechanisms. Five ensembles were tested: proportional damping, weak heterogeneous

Algorithm 1 Adaptive damping-margin estimator

- 1: Input: $\tilde{L}, M, \tilde{D}$, reduced size r , thresholds τ_η, τ_s
- 2: Build $\tilde{L} = M^{-1/2} \tilde{L} M^{-1/2}$ and $\tilde{D} = M^{-1/2} \tilde{D} M^{-1/2}$
- 3: Compute eigenpairs (ν_k, q_k) of $\tilde{L}$
- 4: Compute $\Gamma = Q^T \tilde{D} Q$ and $\hat{\zeta}_k = \Gamma_{kk} / (2\sqrt{\nu_k})$
- 5: Select the critical diagonal mode $c = \arg \min_k \hat{\zeta}_k$
- 6: Compute Δs_c from (22)
- 7: Select low-damping candidates $\mathcal{I}$ by sorting $\hat{\zeta}_k$
- 8: Expand $\mathcal{I}$ with nearby or strongly coupled modes
- 9: **if** $\eta(\mathcal{I}) < \tau_\eta$ and $|\Delta s_c|/|s_0| < \tau_s$ **then**
- 10: Return $\hat{\zeta}_2$ from (23)
- 11: **else**
- 12: Build $L_r = V^T \tilde{L} V$, $D_r = V^T \tilde{D} V$
- 13: Solve $(s^2 I_r + s D_r + L_r) w = 0$
- 14: Return the smallest damping ratio of the reduced roots
- 15: **end if**

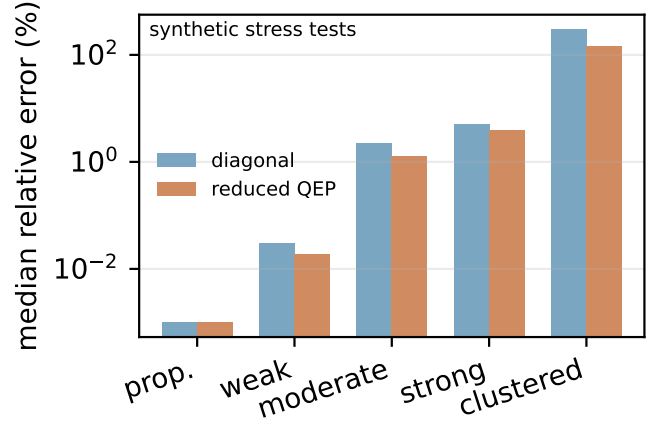


Fig. 2. Synthetic stress-test errors. The diagonal all-mode estimate is excellent in proportional and weak-damping cases, but not robust to strong non-proportional damping or clustered modal frequencies. The reduced QEP is a correction for those regimes, not a replacement for full validation.

damping, moderate heterogeneous damping, strong heterogeneous damping, and near-degenerate clustered modes. For each case the diagonal estimate, the second-order correction, and damping-selected reduced QEPs were compared against the full $2n \times 2n$ eigensolve.

Fig. 2 summarizes the median relative errors for the diagonal and reduced-QEP baselines. The same synthetic audit summarized in Table II shows that the second-order correction reduced the weak-IBR median error from 0.038% to 0.007%; in the underlying run, the 95th-percentile error also fell from 0.881% to 0.163%. In mild and strong heterogeneity, it improved the median error, but it did not reliably beat a larger reduced QEP and could worsen tail errors. This is exactly why the correction is embedded in a diagnostic workflow rather than used as a universal replacement for eigensolve-based validation.

This stress test supports the estimator's logic. The contribution is not "always use the second-order formula" and not "the diagonal formula is always enough." The contribution is the interpretable correction plus the diagnostic boundary between

TABLE II

SYNTHETIC STRESS-TEST SUMMARY. VALUES ARE MEDIAN RELATIVE ERRORS AGAINST THE FULL FIRST-ORDER EIGENSOLVE. THE CLUSTERED CASE IS INTENTIONALLY PATHOLOGICAL AND EXPOSES THE FAILURE MECHANISM.

Regime	Diagonal	2nd order	QEP
Proportional damping	≈ 0	≈ 0	≈ 0
Weak heterogeneity	0.038%	0.007%	0.004–0.024%
Moderate heterogeneity	2.53%	0.65%	0.22–1.51%
Strong heterogeneity	5.16%	1.46%	1.24–3.90%
Clustered frequencies	308%	74%	9.8–160%

diagonal-safe, second-order-safe, and QEP-needed regimes.

E. Failure Mechanism of the Diagonal Screen

The reason for the regime dependence is visible in modal coordinates. Write

$$\Gamma = \Gamma_d + \Gamma_o, \quad (38)$$

where $\Gamma_d = \text{diag}(\delta_k)$ and Γ_o contains the off-diagonal modal damping. If $\Gamma_o = 0$, the QEP decouples. If Γ_o is small and modal frequencies are well separated, the pole perturbation induced by off-diagonal damping is small. If two modes are close, however, the same off-diagonal entry can rotate the critical subspace. A scalar estimate based on one unperturbed q_k then misses the actual least-damped combination of modes.

This mechanism explains why a single error number is not enough. A diagonal formula can be nearly exact in weak damping and badly wrong in clustered modes. A reduced QEP can improve clustered cases, but it is not magic: if the selected subspace omits a coupled neighbor, the reduced model still fails. The estimator must therefore report its own diagnostic label, not only a number.

F. Reduced PLL Check: Network Versus Control

To test whether the perturbation idea survives beyond the clean second-order surrogate, a reduced seven-state grid-following inverter model was evaluated. The three-node system contains one synchronous generator, one grid-forming inverter, and one grid-following inverter with PLL memory. The state vector is

$$x = [\theta_0, \theta_{\text{pll}}, \theta_2, \omega_0, \omega_{\text{pll}}, \omega_2, z_{\text{pll}}]^T, \quad (39)$$

where z_{pll} is the PLL integral state. The zero-order model sets the PLL memory coupling to zero, so the inverter behaves as a memory-free damping channel. Standard nonsymmetric eigenvalue perturbation then corrects the critical pole of this zero-order matrix.

In the canonical reduced case, the full critical pole is approximately $-0.034236 \pm j1.177983$. The zero-order pole is $-0.034827 + j1.178430$, and the second-order correction moves it to $-0.034154 + j1.177846$, reducing the complex-pole distance from 7.41×10^{-4} to 1.60×10^{-4} . Over 399 stable randomized reduced-PLL cases, the median damping-margin error falls from 2.16% at order zero to 0.09% at order two, while the 95th-percentile error falls from 29.2% to 12.9%.

TABLE III

ANDES IEEE39 BASELINE SIMULATION SUMMARY.

Quantity	Value
Buses / lines	39 / 46
Synchronous machines	10 GENROU
IBR dynamic devices	0 REGCP1/REGF1/PLL1
Linearized eigenvalues	160
Oscillatory eigenvalues	64
Positive-real eigenvalues	9 non-oscillatory
Least-damped stable pole	$-1.3460 \pm j8.6115$
Damping ratio / frequency	0.1544 / 1.3706 Hz

This result supports the network-vs-control interpretation: the zero-order margin is the electromechanical network prediction, while the second-order term measures how PLL memory shifts the critical pole. It is not a substitute for ANDES validation. The model is reduced and deliberately transparent; current loops, limiters, algebraic network constraints, and vendor-grade IBR models remain outside this test.

V. ANDES IEEE 39-BUS SIMULATIONS

A. Baseline Synchronous Case

The first IEEE 39-bus experiment was run in ANDES 2.0.0 using the packaged `ieee39_full.xlsx` case. The case contains 39 buses, 46 lines, 10 GENROU machines, 10 IEEEEST/PSS devices, and no active REGCA1, REGCP1, REGF1, REGF2, or PLL1 devices. Thus it is a synchronous-generator benchmark, not a full IBR benchmark. Power flow and eigenanalysis completed. The linearized model has 160 eigenvalues, 64 oscillatory eigenvalues, two zeros, and nine positive-real non-oscillatory modes. The least-damped stable oscillatory pair is

$$s_* = -1.3460 \pm j8.6115, \quad \zeta_* = 0.1544, \quad (40)$$

with oscillation frequency 1.3706 Hz. The positive-real non-oscillatory modes are reported as a case-quality warning; they do not support a final practical stability claim.

B. Positive-Mode Diagnostic

A participation-factor audit was run to identify the source of those positive-real modes. The dominant positive modes participate primarily in TGOV1N, IEEEEX1/IEEEEST, and internal GENROU transient-voltage states rather than in electromechanical angle-frequency pairs. A control experiment removed model families from the packaged case. Removing only TGOV1N reduced the count from nine positive modes to four, while removing only IEEEEST removed all positive-real modes; the least-damped oscillatory pair remained essentially unchanged at approximately $\zeta = 0.1545$ and 1.3705 Hz. Thus the positive modes appear to be a stabilizer-model artifact in the packaged dynamic case, not a network oscillation claim. The baseline is therefore useful for reproducible extraction tests, but the manuscript should not cite it as evidence of a physically unstable New England benchmark.

TABLE IV
ANDES DIAGNOSTIC SIMULATIONS FOR POSITIVE-REAL MODES IN THE
PACKAGED IEEE39 DYNAMIC CASE.

Variant	Removed	Pos. modes	ζ_{osc}
Base	—	9	0.1544
No governor	TGOV1N	4	0.0568
No PSS	IEEEEST	0	0.1545
No exciter/PSS	IEEEEX1, IEEEEST	0	0.1538
GENROU only	Controls	0	0.0606

TABLE V
IEEE39-DERIVED SURROGATE ESTIMATOR ERRORS. VALUES ARE
RELATIVE ERROR AGAINST THE FULL QEP OF THE KRON-REDUCED
SURROGATE, NOT FULL IBR ANDES ERRORS.

Estimator	Median	95th pct.	Mean
Diagonal all-mode	0.468%	24.15%	4.18%
Second order	0.032%	9.91%	1.75%
Reduced QEP ($r = 6$)	0.037%	5.44%	1.09%
Adaptive	0.031%	7.43%	1.63%

C. IEEE39-Derived Kron Surrogate

To exercise the proposed estimator on the IEEE 39 topology before a full IBR dynamic case is built, the solved network was converted into a generator-bus surrogate. The active-power stiffness Laplacian was assembled from the solved line series admittances and bus angles, then Kron-reduced from 39 buses to the 10 generator buses. IBR-like scenarios were generated by reducing effective inertia and introducing heterogeneous local damping at a penetration level ρ . The reference for this surrogate test is the full QEP of the Kron-reduced second-order model, not the full ANDES DAE.

Table V supports the same mechanism as the random ensembles: the second-order correction reduces the median error of the diagonal screen by about one order of magnitude, but the reduced QEP has the better tail. The adaptive workflow is therefore still necessary. In the representative $\rho = 0.6$ scenario, increasing local damping by 10% gives the largest margin improvement at generator bus 30 ($\Delta\zeta = 0.0112$), followed by buses 38 and 36. For line reinforcements, strengthening line 25–37 gives the largest margin increase ($\Delta\zeta = 0.00631$), while strengthening lines 2–3, 25–26, 1–2, 3–4, and 1–39 reduces the damping margin even though the critical modal stiffness increases. This is a damping-aware Braess-like reversal in the surrogate, not yet a full ANDES IBR topology claim.

D. SG-to-GFL/GFM Replacement Cases

To close the gap between the packaged synchronous benchmark and a true IBR benchmark, a case-generation script was added for direct SG-to-IBR replacement in ANDES. The static PV/Slack devices are retained so that the original power-flow dispatch is preserved. For each converted generator, the associated GENROU, governor, exciter, and PSS rows are removed from the derived case. Grid-following replacements use REGCP1, REECA1, REPCA1, and PLL1; grid-forming replacements use REGF1. The slack generator is kept synchronous in all pilot cases.

Table VI gives the first full-ANDES construction result, and Fig. 3 shows the exact bus-level replacement map for the stable Mix60 pilot. All five derived cases load, solve power flow, and run eigenanalysis. Four low-penetration pilot cases retain positive-real modes under the uncalibrated generic controller parameters; the 60% mixed GFL/GFM case has no positive-real eigenvalues under the same tolerance. The stable oscillatory damping ratios in the table are therefore not used as final validation of the estimator. Their immediate value is methodological: the workflow can now generate reproducible IEEE39 cases with active REGCP1/PLL1 and REGF1 dynamics, and the next validation step is to calibrate controller parameters and extract a physically matched reduced surrogate from the full ANDES state matrix.

The frequency column in Table VI is computed as $|\text{Im}(s)|/(2\pi)$, so the entries near 13.7 Hz are not a rad/s-to-Hz unit error. A participation-factor audit shows that these high-frequency least-damped stable modes are dominated by control or auxiliary states, such as IEEEEST/IEEEEX1, REGF1, PLL1, or BusFreq states, rather than by a low-frequency inter-area electromechanical mode. This is consistent with the central warning of the paper: once IBR and controller states are active, the least-damped stable mode need not be the familiar low-frequency network mode.

The scripts store this gate table, the Mix60 topology diagram, and the supporting JSON/CSV artifacts under `outputs/ieee39_phase0_bridge`. This makes the current empirical status explicit: the paper has reproducible ANDES baseline and SG-to-IBR construction artifacts, plus IEEE39-derived surrogate estimator tests, but it does not yet have the calibrated full-ANDES IBR comparison required for a final Transactions-level empirical claim.

E. Phase-1 Controller Calibration Audit

A first calibration audit was then run on the five derived IBR cases. Ten documented profiles were tested: the original generic parameters; slower PLL gains; softened GFL voltage/reactive controls; softened GFM controls; combined soft profiles; and diagnostic variants that remove retained synchronous-machine IEEEEST/PSS or IEEEEX1+IEEEEST rows. Each profile was evaluated by full ANDES power flow and eigenanalysis, with participation factors recorded for the dominant positive mode and the least-damped stable oscillatory mode.

Table VIII is a negative and useful result. Merely slowing or softening the IBR controllers does not stabilize the low-penetration cases. Diagnostic removal of retained synchronous-machine stabilizer/exciter rows stabilizes GFL20, Mix20, and Mix40, while GFM20 remains unstable even under the diagnostic removal. Thus the remaining benchmark blocker is broader than IBR gain selection: a defensible full-ANDES IBR benchmark must choose or calibrate the retained synchronous-machine controls and the inverter controls together, then validate a reduced surrogate against the full calibrated Jacobian before estimator errors are reported.

IEEE 39-bus Mix60 SG-to-IBR replacement map

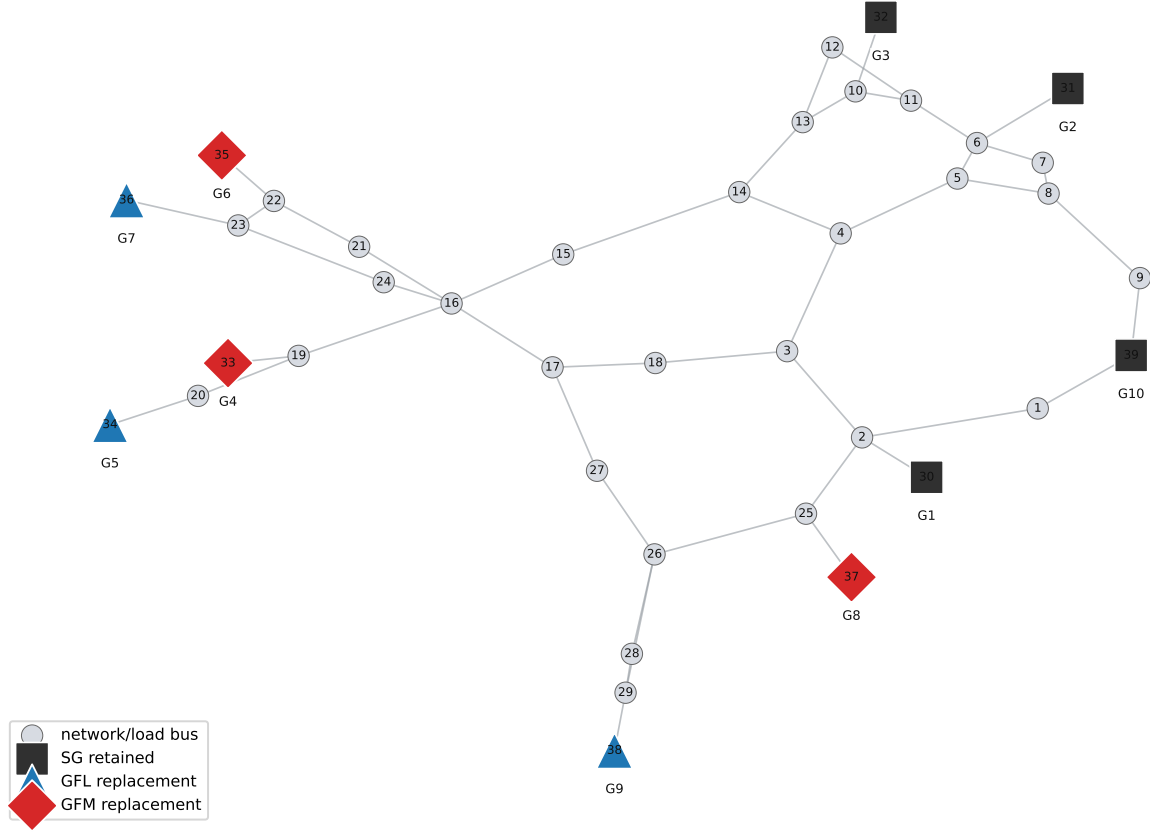


Fig. 3. IEEE 39-bus topology and the Mix60 SG-to-IBR replacement map used in the ANDES pilot. Generators 5, 7, and 9 are replaced by GFL devices at buses 34, 36, and 38; generators 4, 6, and 8 are replaced by GFM devices at buses 33, 35, and 37; generators 1, 2, 3, and 10 remain synchronous at buses 30, 31, 32, and 39. This figure documents the benchmark construction. It is not, by itself, estimator validation.

TABLE VI

FULL ANDES IEEE39 SG-TO-IBR REPLACEMENT PILOT SIMULATIONS. ALL CASES SOLVED POWER FLOW AND EIGENANALYSIS. STABILITY IS ASSESSED FROM THE FULL LINEARIZED DAE WITH GENERIC UNCALIBRATED CONTROLLER PARAMETERS.

Case	GFL gens	GFM gens	GENROU left	Pos. modes	Max $\text{Re}(s)$	Stable	ζ_{osc} / Hz
GFL20	9,8	—	8	1	1.012	no	0.1148 / 13.7277
GFM20	—	9,8	8	6	4.291	no	0.1438 / 1.3910
Mix20	9	8	8	3	1.070	no	0.1370 / 13.7507
Mix40	9,7	8,6	6	2	1.072	no	0.1266 / 13.7856
Mix60	9,7,5	8,6,4	4	0	2.0×10^{-9}	yes	0.1236 / 13.7417

F. Baselines Required for a Fair Test

Four baselines should be reported in every validation table. The first is the homogeneous damping formula, which is valid only when damping is proportional or can be represented by a scalar. The second is a Fiedler-only estimate, which tests whether low-frequency intuition is being confused with damping-margin estimation. The third is a frequency-selected reduced QEP, which tests whether reduction succeeds merely because it is low order or because it selects the correct low-damping subspace. The fourth is the full eigensolve, which is the reference and not a competitor.

These baselines prevent two common mistakes: crediting

the adaptive estimator for a case where the diagonal all-mode formula already works, and crediting a graph-frequency proxy for a case where the mechanism is damping.

VI. PLANNING WITH THE DAMPING MARGIN

The estimator becomes useful when it ranks actions. Let a denote a planning action: conversion of a synchronous generator to an IBR, addition of virtual damping, addition of virtual inertia, or local controller retuning. Define

$$J(a) = \zeta_{\min}(L(a), M(a), D(a)), \quad (41)$$

TABLE VII
PHASE-0 VALIDATION GATES FOR THE FULL ANDES IEEE39 IBR BENCHMARK.

Gate	Status	Evidence and required next step
Baseline ANDES load/PFlow/EIG Positive-mode diagnostic	pass pass-diagnostic	39 buses, 46 lines, and 160 eigenvalues; synchronous extraction benchmark only. Removing IEEEEST/PSS removes all positive-real modes; report as case-quality warning.
SG-to-GFL/GFM case generation Generic IBR small-signal stability	pass partial	Five derived cases solved power flow and eigenanalysis. One of five cases is stable under generic gains; controller calibration is still required.
Physical ANDES-to- (L, M, D) bridge Estimator validation vs. full ANDES IBR poles	blocked blocked	A matched reduced model from the full IBR Jacobian must be validated first. Requires calibrated IBR dynamics and the physical bridge before error claims are made.

TABLE VIII
PHASE-1 CONTROLLER CALIBRATION AUDIT. DIAGNOSTIC PROFILES REMOVE RETAINED SYNCHRONOUS-MACHINE CONTROLS AND ARE NOT FINAL TUNING RECOMMENDATIONS.

Case	Best controller-only profile	Best diagnostic profile	Mode audit
GFL20	Very-soft IBR controls: 1 positive mode, max $\text{Re}(s) = 1.011$.	No PSS plus soft IBR controls: stable, max $\text{Re}(s) = 4.0 \times 10^{-10}$.	Generic positive mode dominated by REECA1; high-frequency stable modes are control/auxiliary dominated.
GFM20	Baseline remains best: 6 positive modes, max $\text{Re}(s) = 4.291$.	No exciter/PSS: still 2 positive modes, max $\text{Re}(s) = 4.286$.	Remaining instability is not cleared by the tested IBR profiles; retained SG/exciter dynamics require separate treatment.
Mix20	Slow PLL: 3 positive modes, max $\text{Re}(s) = 1.066$.	No exciter/PSS: stable, max $\text{Re}(s) = 8.3 \times 10^{-8}$.	The least-damped stable mode near 13.75 Hz is high-frequency auxiliary/control dominated.
Mix40	Slow PLL: 2 positive modes, max $\text{Re}(s) = 1.065$.	No exciter/PSS: stable, max $\text{Re}(s) = 8.9 \times 10^{-8}$.	The stable high-frequency critical mode has leading PLL1 participation.
Mix60	Soft GFL controls: stable, max $\text{Re}(s) = 4.0 \times 10^{-10}$.	No PSS: stable, max $\text{Re}(s) = 2.0 \times 10^{-9}$.	The controller-only stable critical mode is dominated by REGF1 participation.

TABLE IX
PLANNING QUANTITIES AND INTENDED USE.

Quantity	Use
$\zeta_{\min}(a)$	Safety of finite conversion or reinforcement
$\Delta\zeta_{\min}/\text{cost}$	Cost-effective damping or inertia placement
$\partial\zeta_{\min}/\partial w_e$	Local line-reinforcement screening
$\partial\nu_k/\partial w_e$	Frequency mechanism, not final ranking
$q_k(i)^2$	Modal participation diagnostic

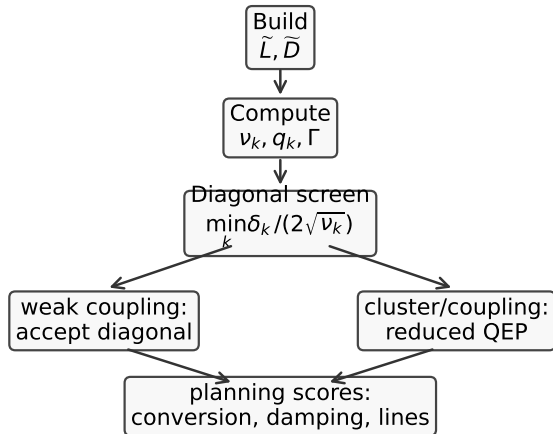


Fig. 4. Workflow of the adaptive estimator. The diagonal region is the first screen; second-order perturbation supplies the cheap correction; the reduced QEP is triggered by clustered low-damping modes or strong off-diagonal modal damping.

and, when costs are available,

$$J_c(a) = \frac{\zeta_{\min}(a) - \zeta_{\min}(0)}{\text{cost}(a)}. \quad (42)$$

The action with highest J is the safest conversion; the action with highest J_c is the most cost-effective improvement.

A. Interpretation of the Planning Score

The score is defined using the post-action margin rather than only a derivative at the current operating point. A derivative

is local. A conversion from synchronous generation to an inverter can be a finite perturbation to inertia, damping, and control states. For small damping additions, a local formula is useful. For finite SG-to-IBR conversion, the estimator should be recomputed after each candidate action.

The distinction between absolute and incremental scores matters. A candidate action can produce a large incremental improvement but still leave the system below the target margin. Conversely, a candidate can yield modest improvement but preserve the margin during a conversion sequence. For decarbonization planning, the safest action can be written as

$$a^* = \arg \min_{a \in \mathcal{A}} [\zeta_* - \zeta_{\min}(a)]_+, \quad (43)$$

with cost normalization used after safety is satisfied.

B. SG-to-IBR Conversion

A single conversion can be modeled at the reduced level by changing the inertia and damping assigned to the candidate bus:

$$M_i \leftarrow M_{\text{IBR}}, \quad D_i \leftarrow D_{\text{IBR}}. \quad (44)$$

The exact mapping depends on whether the replacement is grid-following or grid-forming. A grid-following inverter should not be treated as a perfect nodal damping element in final models; the approximation is a planning surrogate that must be validated against full IBR state-space models. The value of the estimator is that it screens many candidates cheaply before full eigensolve or time-domain analysis.

The baseline comparisons are essential. A Fiedler-participation rule ranks buses by $q_2(i)^2$. A frequency-participation rule ranks buses by participation in the highest-frequency mode. An inertia heuristic ranks buses by current M_i . None of these directly optimizes (9). The proposed rule ranks by post-conversion margin. If a simple proxy matches the exact margin in a benchmark, that result should be reported; if it fails, the failure is the justification for the spectral estimator.

C. Damping and Virtual-Inertia Placement

For small additions ΔD_i or ΔM_i , a first-order placement score can be derived from the diagonal region. If D_i changes while M and L are fixed,

$$\frac{\partial \delta_k}{\partial D_i} = \frac{q_{k,i}^2}{M_i}, \quad (45)$$

and hence

$$\frac{\partial \zeta_k}{\partial D_i} \approx \frac{q_{k,i}^2}{2M_i \sqrt{\nu_k}}. \quad (46)$$

This expression gives a transparent damping-placement score, but it must be applied to the mode that determines $\zeta_{\min}$ and corrected when modal coupling is strong. Virtual inertia is more delicate because it changes the normalization, modal frequencies, and damping geometry simultaneously. Therefore, inertia placement should be evaluated by recomputing the adaptive estimator after the candidate action rather than by relying solely on a closed-form scalar score.

For infinitesimal inertia actions, however, Appendix B gives the closed-form score

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial M_i} + \frac{\delta_k q_{k,i}^2}{4M_i \sqrt{\nu_k}}, \quad (47)$$

with $\partial \delta_k / \partial M_i$ in (78). Unlike (46), this derivative is sign-indefinite: inertia changes both the graph frequency and the modal damping geometry.

A useful sign test follows when the local damping-per-inertia ratio $\bar{d}_i = D_i/M_i$ is held fixed during an infinitesimal inertia change. This removes the trivial effect in which fixed physical damping is diluted by larger inertia. Let

$$G_i = \frac{\partial \tilde{L}}{\partial M_i} = -\frac{E_i \tilde{L} + \tilde{L} E_i}{2M_i}, \quad (48)$$

where $E_i = e_i e_i^T$. For a simple mode k ,

$$\frac{\partial \nu_k}{\partial M_i} = -\frac{\nu_k q_{k,i}^2}{M_i}, \quad (49)$$

and the modal-damping rotation term is

$$\left(\frac{\partial \delta_k}{\partial M_i} \right)_{\text{rot}} = 2 \sum_{\ell \neq k} \frac{(q_\ell^T G_i q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (50)$$

TABLE X
INERTIA-REVERSAL MONTE CARLO CHECK. A REVERSAL MEANS THAT INCREASING M_i LOWERS THE DAMPING MARGIN UNDER A FINITE-DIFFERENCE PERTURBATION.

Assumption	Tests	Full rev.	Diag+full rev.
Fixed D_i/M_i	16229	7694	5447
Fixed D_i	16229	11054	8425

Thus the diagonal modal damping-ratio derivative is

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \left(\frac{\partial \delta_k}{\partial M_i} \right)_{\text{rot}} + \frac{\delta_k q_{k,i}^2}{4M_i \sqrt{\nu_k}}. \quad (51)$$

The second term in (51) is positive, but the rotation term can be negative. Therefore, increasing inertia can reduce the diagonal damping-ratio screen whenever

$$\left(\frac{\partial \delta_k}{\partial M_i} \right)_{\text{rot}} < -\frac{\delta_k q_{k,i}^2}{2M_i}. \quad (52)$$

This is not a claim that inertia is harmful in general. It is a sign condition showing that, under non-proportional damping, inertia placement is not monotone in the damping margin.

Table X reports a reproducible random-graph audit. The fixed- D_i case is included as a control because it contains direct damping dilution. The fixed- D_i/M_i case is the nontrivial check: in the strongest five-node example, (51) predicts $\partial \zeta_k / \partial M_i = -0.0373$, the matched-mode finite difference gives -0.0397 , and the full-QEP damping-margin finite difference is also negative (-0.106). The larger full-QEP derivative is not a contradiction. Equation (51) is a diagonal matched-mode sensitivity, while the full-QEP margin also includes off-diagonal modal coupling and possible movement of the identity of the critical mode. In this example, the coupled system amplifies the same sign reversal detected by the diagonal formula. This supports the mechanism but remains a reduced-model result; it has not yet been validated on calibrated ANDES IBR dynamics.

D. Progressive Conversion

A practical transition is not a single conversion; it is a sequence. Let $\mathcal{C}$ be the set of synchronous generators still eligible for conversion. At step m , define

$$i_m = \arg \max_{i \in \mathcal{C}} \zeta_{\min}(M^{(m,i)}, D^{(m,i)}, L^{(m)}), \quad (53)$$

where $(M^{(m,i)}, D^{(m,i)}, L^{(m)})$ is the model after converting candidate i on top of the previous $m-1$ conversions. This greedy rule should be compared against a fixed merit order, random order, inertia order, and modal-participation order.

Progressive conversion is important because single-action ranking can overstate robustness. A bus that is safe to convert first may become unsafe after another conversion changes the critical mode. The benchmark should report the full margin trajectory versus IBR penetration, not only the first selected bus. If the greedy rule fails at a certain penetration level, that failure is a useful result: it identifies where local planning must be replaced by combinatorial or robust optimization.

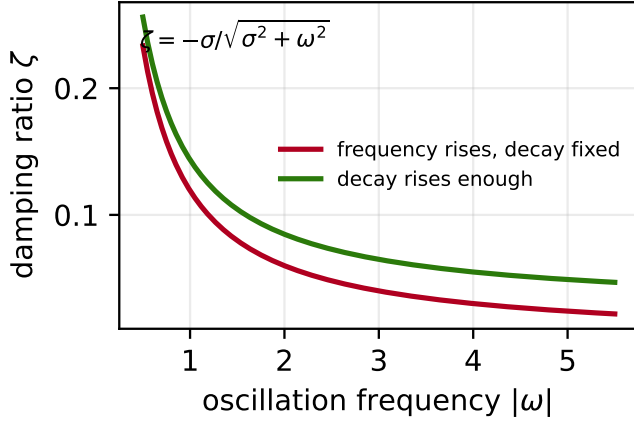


Fig. 5. Frequency–damping tradeoff. If a line reinforcement increases oscillation frequency without increasing decay rate enough, damping ratio decreases. This is not a topological Braess paradox; it is the definition of damping ratio.

VII. DAMPING-AWARE LINE REINFORCEMENT

The old line map is the Rayleigh sensitivity of the inertial-Laplacian eigenvalue:

$$\frac{\partial \nu_k}{\partial w_{ij}} = \left(\frac{q_{k,i}}{\sqrt{M_i}} - \frac{q_{k,j}}{\sqrt{M_j}} \right)^2. \quad (54)$$

This formula is useful, but it answers the wrong planning question if used alone. It says which line raises modal stiffness or frequency. It does not say which line raises damping margin. Equivalently, the unnormalized Laplacian perturbation is $\partial L / \partial w_{ij} = (e_i - e_j)(e_i - e_j)^T$, while the perturbation used in modal formulas is the inertia-normalized matrix

$$\frac{\partial \tilde{L}}{\partial w_{ij}} = M^{-1/2} (e_i - e_j)(e_i - e_j)^T M^{-1/2}. \quad (55)$$

The reason is immediate from the diagonal formula:

$$\zeta_k = \frac{\delta_k}{2\sqrt{\nu_k}}. \quad (56)$$

Differentiating with respect to a line weight w_e gives

$$\frac{\partial \zeta_k}{\partial w_e} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial w_e} - \frac{\delta_k}{4\nu_k^{3/2}} \frac{\partial \nu_k}{\partial w_e}. \quad (57)$$

The second term is negative when reinforcement increases ν_k . Thus, a line can raise frequency while reducing damping ratio if the first term does not compensate. Fig. 5 illustrates the same mechanism in pole coordinates: increasing $|\omega|$ while holding $-\sigma$ fixed lowers $\zeta = -\sigma/\sqrt{\sigma^2 + \omega^2}$.

For a simple eigenvalue of $\tilde{L}$, the derivative of modal damping is

$$\frac{\partial \delta_k}{\partial w_e} = q_k^T \frac{\partial \tilde{D}}{\partial w_e} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T \frac{\partial \tilde{L}}{\partial w_e} q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (58)$$

If line reinforcement affects only L , the first term is zero, but the second term can be nonzero because the eigenvectors rotate relative to the damping matrix. This is the precise

mechanism by which non-proportional damping changes line-ranking decisions.

For the exact QEP, let s be a simple critical root with right and left eigenvectors v and y :

$$P(s)v = 0, \quad y^* P(s) = 0. \quad (59)$$

The normalization of v and y is arbitrary because it cancels between numerator and denominator in the sensitivity expression; the only requirement is $y^* P'(s)v \neq 0$ for a simple root. For a scalar parameter ρ ,

$$\frac{ds}{d\rho} = -\frac{y^* \left(s \frac{\partial \tilde{D}}{\partial \rho} + \frac{\partial \tilde{L}}{\partial \rho} \right) v}{y^* (2sI + \tilde{D}) v}, \quad (60)$$

assuming the coefficient of s^2 is fixed in the chosen coordinates. If $s = \sigma + j\omega$ and $ds/d\rho = a + jb$, then

$$\frac{d\zeta}{d\rho} = \frac{\sigma\omega b - \omega^2 a}{|s|^3}. \quad (61)$$

Equations (60)–(61) define the damping-aware line map. The frequency map (54) remains useful for mechanism interpretation, but the ranking variable is $\partial \zeta_{\min} / \partial w_e$ or a finite post-action $\Delta \zeta_{\min}$.

A. Finite-Difference Validation

Before using (60) operationally, the analytic sensitivity should be checked against finite differences. For edge $e = (i, j)$ with base weight w_e , define

$$\Delta_e^{\text{FD}}(\epsilon) = \frac{\zeta_{\min}(w_e(1 + \epsilon)) - \zeta_{\min}(w_e)}{\epsilon w_e}. \quad (62)$$

The perturbation should preserve Laplacian row sums:

$$\Delta L_e = \epsilon w_e (e_i - e_j)(e_i - e_j)^T. \quad (63)$$

The analytic sensitivity should converge to Δ_e^{FD} as ϵ decreases until roundoff dominates. If the identity of the critical eigenvalue changes during the perturbation, the line should be flagged as a nonsmooth ranking point rather than forced into a smooth derivative.

B. Why Frequency-Only Ranking Can Fail

The frequency map (54) is nonnegative in the scalar lossless model:

$$\frac{\partial \nu_k}{\partial w_e} \geq 0. \quad (64)$$

Thus reinforcement increases or preserves modal stiffness. The mistake is to equate stiffness with damping margin. For a lightly damped pole, $\omega \approx \sqrt{\nu_k}$ while σ is set by damping and modal rotation. If ω increases while σ is fixed, ζ decreases. If eigenvector rotation moves the mode toward lower-damped buses, σ can also become less negative. The corrected map therefore decomposes reinforcement into a frequency term and a modal-damping-rotation term.

VIII. BENCHMARK VALIDATION PROTOCOL

The decisive validation target is the IEEE 39-bus New England system in ANDES [20]. A final submission should not rely only on random graphs or low-order examples. The intended validation pipeline is as follows.

A. System Setup

The base case should be initialized with a solved power flow. Synchronous machines should use documented dynamic models available in ANDES, such as GENROU or the closest supported equivalent. IBR penetration should be introduced by replacing a fraction ρ of synchronous capacity with documented inverter models. Grid-following replacements should include PLL dynamics where available. Grid-forming replacements should use droop or virtual-synchronous-machine models where available. Controller gains should be documented and should not be hand-tuned to make the proposed method look better.

B. Experiments

Experiment A compares margin estimators. For each penetration level ρ , compute the full small-signal poles using ANDES, then compare the homogeneous formula, Fiedler-only formula, diagonal all-mode estimate, and adaptive reduced QEP.

Experiment B tests the coupling diagnostic. The expected outcome is not that the diagonal estimate always fails or always succeeds. The expected outcome is that the diagnostic predicts when it can be trusted.

Experiment C ranks SG-to-IBR conversions. Each candidate conversion is evaluated by the adaptive estimator and by full eigensolve. Baselines include Fiedler participation, highest-frequency participation, and inertia-only heuristics.

Experiment D tests multi-node conversion. Greedy conversion by post-action damping margin is compared against random conversion and heuristic merit orders. Failures must be reported, especially if interactions make the greedy rule unreliable.

Experiment E evaluates line reinforcement. Rank lines by $\partial\nu/\partial w$, finite-difference $\Delta\zeta_{\min}$, and analytic QEP sensitivity. The key test is whether frequency-only reinforcement disagrees with damping-aware reinforcement and whether the analytic sensitivity predicts the full eigensolve.

C. Extracting Reduced Quantities From ANDES

The ANDES bridge must be documented carefully. The full ANDES state matrix contains device states, algebraic variables, controllers, and possibly load or network dynamics. The reduced spectral model requires L , M , and D or their best physically justified equivalents.

The extraction should proceed in two levels. Level 1 constructs the second-order surrogate. The active-power Jacobian with respect to bus angles gives the stiffness matrix corresponding to L under the assumptions of (1). Synchronous-machine inertia and virtual inertia terms define M_i . Mechanical damping, droop, and equivalent local damping define D_i only when the mapping is physically meaningful. Level 2 compares that surrogate against the full ANDES linearization with IBR controller states. Failure at Level 2 would not invalidate the mathematics of the estimator; it would delimit the regime where a scalar L, M, D surrogate is insufficient.

TABLE XI
PRE-REGISTERED ANDES VALIDATION MATRIX.

Experiment	Metric
Estimator accuracy	Median and 95th-percentile error vs full eigensolve
Coupling trigger	Precision/recall of diagonal-failure detection
Conversion ranking	Top-1 and top-3 agreement with full eigensolve
Multi-conversion	Margin trajectory vs penetration level
Line sensitivity	Correlation and top- k overlap with finite differences
Surrogate fidelity	Scalar model poles vs full ANDES IBR poles

D. Success Criteria

The method should be judged by pre-registered criteria:

- median and 95th-percentile error of margin estimates,
- fraction of cases where the adaptive estimator identifies the correct critical mode or cluster,
- top- k accuracy of conversion ranking,
- agreement between analytic and finite-difference line sensitivities,
- failure cases and their mechanisms.

If the adaptive estimator fails at high IBR penetration, that result should be included. The value of the paper is a calibrated method, not a universal claim.

E. Decision Gates for Claim Strength

The final benchmark should assign each contribution a claim level before submission. The highest level is an empirical contribution: the method is validated against full ANDES poles over the stated operating range and beats the registered baselines. The middle level is a theoretical or diagnostic contribution: the mathematics is correct and the mechanism is visible, but benchmark advantage is narrow or case-dependent. The lowest level is a cautionary result: the proposed idea identifies a failure mode of simpler reasoning, but it is not strong enough to headline the paper.

For C1, the empirical level requires that the modal points in the (ν, δ) plane predict safe and unsafe cases in the commuting or weakly coupled regime, and that the same plot visually explains at least one non-proportional failure. If the plot is only exact in toy proportional cases, C1 should be framed as theory background plus visualization, not as a benchmark contribution.

For C2, the empirical level requires that the adaptive estimator reduce error or computational burden relative to full eigensolve and relative to the diagonal screen in the cases where the coupling diagnostic fires. If the diagonal all-mode formula is already accurate on every IEEE 39 case, the reduced QEP should be demoted to a safeguard for degenerate or artificially clustered modes. This would still be a valid result, but it would change the title and contribution ordering.

For C3, the empirical level requires low regret, not only high correlation. A conversion ranking that is usually correlated with the full eigensolve but occasionally chooses an action with a much lower margin is not operationally acceptable. The

TABLE XII
CLAIM LEVELS USED AFTER BENCHMARK VALIDATION.

Level	Meaning
Empirical contribution	Verified against full benchmark poles and stronger than registered baselines
Diagnostic contribution	Mathematically correct and mechanism-clear, but benchmark advantage is case-dependent
Cautionary result	Useful for avoiding a wrong inference, but not a standalone planning method
Rejected claim	Control comparison or mechanism check fails

correct fallback claim would be that the estimator provides a screening list for full eigensolve, not a standalone planner.

For C4, the empirical level requires two checks. First, analytic sensitivity must match finite differences when the critical pole is smooth. Second, damping-aware ranking must differ materially from frequency-only ranking in at least some realistic operating points. If both rankings agree on IEEE 39, the result should be stated as a theoretical correction that prevents a known category error rather than as evidence of a frequent planning reversal.

F. Negative Controls

The benchmark should include controls that are expected to fail. These controls are not straw men; they are the reason the proposed method is needed. A Fiedler-only estimate should fail whenever the damping-critical mode is not the lowest nonzero graph frequency. A homogeneous damping formula should fail when IBR conversion creates large variation in D_i/M_i . A frequency-only line map should fail whenever reinforcement raises ν_k while modal damping does not rise enough to preserve ζ_k . A low-frequency QEP should fail when the least damped mode is outside the retained frequency band.

Reporting these controls prevents a common source of false positives. If the proposed estimator beats only an obviously wrong control but not the diagonal all-mode screen, then the central claim is not the reduced QEP. If the damping-aware line map beats frequency sensitivity but not finite-difference full eigensolve, then the analytic implementation is wrong or the pole is nonsmooth. Each negative control therefore has a paired diagnostic that identifies whether the failure is caused by the baseline assumption, the proposed approximation, or the reduced surrogate itself.

IX. CLAIM CALIBRATION AND EVIDENCE MAP

The proposed paper has to survive a strict novelty and mechanism audit. This section states how each contribution should be defended and what evidence would be required in a final submission.

A. C1 Is a Region Framework, Not a New Scalar Formula

The formula $\zeta_k = \delta_k/(2\sqrt{\nu_k})$ in the commuting case is a standard modal result. The contribution is not the formula. The contribution is the use of the (ν, δ) plane as a planning coordinate for heterogeneous IBR damping, together with the statement that this coordinate is exact only in the commuting

case and diagnostic otherwise. The consensus-region analogy is useful because it changes the question from “which graph eigenvalue is critical?” to “which modal points lie outside the admissible damping region?” This avoids the common mistake of treating λ_2 as a damping margin.

Fig. 1 therefore includes two visual checks. The commuting case shows where the region is an exact margin certificate. The non-proportional panel shows the failure mode: a diagonal modal point can lie above the target curve while the coupled QEP pole falls below the margin. This makes C1 a boundary-aware framework rather than a standalone novelty claim.

B. C2 Is a Mode-Selection Rule, Not Generic Model Reduction

Reduced QEPs are well known. The estimator’s claim is the selection logic. Low-frequency truncation is natural in inter-area oscillation analysis, but the damping margin is the minimum damping ratio over all oscillatory modes. A high-frequency mode can be less damped than a Fiedler mode if δ_k is small enough. Conversely, a low-frequency mode can be safe if its modal damping is large. The reduced model should therefore be built from low estimated damping and strong modal coupling, not from frequency alone.

The evidence needed for this claim is an ablation table. Let ζ_{full} be the reference. Compare:

$$e_{\text{hom}} = |\zeta_{\text{hom}} - \zeta_{\text{full}}|/\zeta_{\text{full}}, \quad (65)$$

$$e_{\text{Fiedler}} = |\zeta_2 - \zeta_{\text{full}}|/\zeta_{\text{full}}, \quad (66)$$

$$e_{\text{freqQEP}} = |\zeta_{\text{lowfreqQEP}} - \zeta_{\text{full}}|/\zeta_{\text{full}}, \quad (67)$$

$$e_{\text{adapt}} = |\zeta_{\text{adapt}} - \zeta_{\text{full}}|/\zeta_{\text{full}}. \quad (68)$$

The adaptive estimator is justified only if e_{adapt} improves the cases where simple baselines fail and if the diagnostic explains the improvement.

C. C3 Is a Planning Rule With a Full-Eigensolve Reference

The conversion and placement contribution should be judged by ranking accuracy, not only by correlation. A high correlation can hide top-action mistakes. Let $r_{\text{est}}(i)$ and $r_{\text{full}}(i)$ be the estimated and true ranks of candidate action i . Useful metrics include top-1 hit rate, top-3 overlap, Kendall rank correlation, and regret:

$$\text{regret} = \zeta_{\min}(a_{\text{full}}^*) - \zeta_{\min}(a_{\text{est}}^*). \quad (69)$$

Regret is especially important for planning. If several actions are nearly tied, a wrong top-1 selection may have negligible regret. If the wrong action collapses the margin, the ranking method is not acceptable even with high average correlation.

D. C4 Is a Damping Sensitivity, Not a Braess Claim

The line-reinforcement contribution is easiest to overstate. The paper should not say that reinforcement creates a new dynamic Braess paradox unless a topological feedback mechanism is demonstrated. The defensible claim is simpler and stronger: frequency-only line ranking can be wrong because damping ratio depends on both decay and frequency. The

TABLE XIII
CONTRIBUTION STATUS AND REQUIRED VALIDATION.

Component	What is established	What is not established	Required next check
Damping region	Exact for commuting damping; first-order screen for weak non-proportional damping	Novelty of the region language for IBR planning	Literature search and IEEE 39 modal plots
Adaptive estimator	Synthetic tests separate diagonal-safe and QEP-needed regimes	Universal diagnostic threshold	ANDES full eigensolve comparison
Conversion ranking	Planning score is well-defined by post-action $\zeta_{\min}$	Multi-node interaction robustness	Greedy conversion sweep on IEEE 39
Damping-aware line map	Mechanism and QEP sensitivity formulas are derived	Analytic implementation vs finite differences	Line perturbation sensitivity in ANDES

Algorithm 2 Damping-aware line ranking

- 1: Input: L, M, D , edge set $\mathcal{E}$, perturbation size ϵ
- 2: Compute $\zeta_{\min}$ and the critical pole s
- 3: **for** each edge $e = (i, j) \in \mathcal{E}$ **do**
- 4: Build $\partial L / \partial w_e = (e_i - e_j)(e_i - e_j)^T$
- 5: Evaluate analytic ds/dw_e from (60)
- 6: Convert ds/dw_e to $d\zeta/dw_e$ using (61)
- 7: Optionally compute finite-difference $\Delta_e^{\text{FD}}(\epsilon)$
- 8: **end for**
- 9: Rank edges by $d\zeta_{\min}/dw_e$ or finite $\Delta\zeta_{\min}$
- 10: Report $\partial\nu/\partial w_e$ only as a mechanism diagnostic

Algorithm 3 Greedy SG-to-IBR conversion screening

- 1: Input: base model (L, M, D) , candidates $\mathcal{C}$, target penetration
- 2: Initialize converted set $\mathcal{S} = \emptyset$
- 3: **while** penetration target not reached **do**
- 4: **for** each $i \in \mathcal{C} \setminus \mathcal{S}$ **do**
- 5: Apply candidate conversion i to obtain (L_i, M_i, D_i)
- 6: Estimate $\hat{\zeta}_i$ with Algorithm 1
- 7: **end for**
- 8: Convert $i^* = \arg \max_i \hat{\zeta}_i$
- 9: Append i^* to $\mathcal{S}$ and recompute the operating point if required
- 10: **end while**
- 11: Validate the selected sequence against the full eigensolve

exact QEP sensitivity gives the derivative of the pole, and (61) converts it to a damping-ratio derivative.

The final line-reinforcement experiment should report three lists for each operating point: top lines by $\partial\nu/\partial w$, top lines by analytic $\partial\zeta/\partial w$, and top lines by finite-difference $\Delta\zeta$. A disagreement between the first and the latter two is the evidence for C4. Agreement between analytic sensitivity and finite difference is the evidence that the formula is implemented correctly.

X. DISCUSSION

The proposed framework has three important limits. First, the scalar swing-equation Laplacian is not a full dq-domain inverter-network model. It captures the electromechanical and stiffness geometry used by many spectral power-network studies, but high-order inverter states, current loops, limiters, and

TABLE XIV
EVIDENCE REQUIRED BEFORE FINAL SUBMISSION.

Claim	Required evidence
C1 region	Modal plots showing safe/unsafe boundary and failures under coupling
C2 estimator	Ablation against homogeneous, Fiedler, frequency-QEP, and full eigensolve
C3 planning	Top- k ranking, regret, and multi-conversion trajectory
C4 line map	Analytic-vs-finite-difference sensitivity and disagreement with frequency-only ranking

dynamic line models can change the relevant modes. The ANDES validation stage is therefore not optional.

Second, the second-order correction is not a universal replacement for a reduced or full eigensolve. It improves the diagonal screen in weak-to-moderate non-proportional regimes and gives an interpretable coupling decomposition, but it can worsen tail errors in strongly coupled or near-degenerate cases. Those cases are exactly why the reduced QEP remains in the workflow.

Third, the line-reinforcement result should not be described as a dynamic Braess paradox unless a topological mechanism is demonstrated. If reinforcement raises frequency and lowers damping ratio because decay does not increase enough, the mechanism is ordinary damping-ratio geometry. That distinction matters. The paper should say that frequency-only reinforcement can be damping-counterproductive, not that every such case is a paradox.

The H-infinity disturbance view and switched-topology Lyapunov view are valuable extensions, but they should not be central in this manuscript. The H-infinity view asks which disturbance shapes excite the reduced system. It is complementary to damping margin but risks reviving poorly defined “weak pathway” claims if not controlled carefully. The switched-topology view requires common or multiple Lyapunov functions and dwell-time assumptions; it is a larger thesis direction rather than a clean first paper.

A. Originality Boundaries

The paper’s novelty boundary should be stated plainly. Modal swing-equation formulas under proportional damping, Rayleigh eigenvalue sensitivity, QEP sensitivity, broad stability-region ideas, and eigenvalue perturbation are known.

The candidate contribution is not new mathematics in isolation. It is a power-systems contribution: use a closed-form second-order perturbation of the critical damping-margin pole to separate the network prediction from inverter-control coupling, and embed that correction in a diagnostic workflow with QEP fallback.

This boundary matters for peer review. A reviewer can object if the paper appears to claim that $\zeta_k = \delta_k/(2\sqrt{\nu_k})$ or generic QEP sensitivity is new. The defensible claim is that the paper uses the diagonal relation as the zero-order network margin, derives the second-order control-coupling correction for the critical margin pole, and shows when this cheap correction should yield to a reduced QEP. Originality should be judged by the formulation, the control comparisons, and the benchmark results, not by whether every mathematical identity is new.

In final form, each contribution should therefore be labeled as follows. The central contribution is the second-order margin correction and the network-vs-control interpretation, verified first in the surrogate QEP and reduced PLL model. The diagnostic workflow is original only if its trigger reliably separates diagonal-safe, second-order-safe, and QEP-needed cases. Planning and line-reinforcement results are applications; they become headline contributions only if ANDES validation shows lower regret or materially different line rankings than registered controls.

B. Falsification Criteria

Three outcomes would weaken the paper and should be reported if observed. First, if the second-order correction is no more accurate than the diagonal all-mode formula on IEEE 39 across relevant IBR penetrations, it should be presented only as an interpretive decomposition. Second, if the reduced PLL or scalar L, M, D surrogate does not track full ANDES IBR poles, the contribution should be reframed as a reduced-model screening method rather than a full IBR stability estimator. Third, if damping-aware line ranking does not differ materially from frequency-only ranking on the benchmark, C4 should be reduced to a theoretical caution rather than a headline application.

These criteria make the manuscript stronger. They state before the benchmark what evidence would consolidate, weaken, or refocus the claims.

C. Final Claim-Readiness Audit

The current repository includes a final claim audit generated from the versioned validation outputs. Table XV summarizes the status at the close of this draft. The central mathematical and reduced-model claims are supported; the full calibrated ANDES IBR benchmark claim is not yet supported. This distinction is the intended submission boundary of the manuscript.

XI. CONCLUSION

This paper proposed a second-order spectral perturbation framework for damping-margin estimation in inverter-dominated grids. The zero-order object is the pair (ν_k, δ_k) :

inertial-Laplacian stiffness and modal damping. In non-proportional IBR regimes, off-diagonal modal damping moves the critical pole away from this diagonal network prediction. The closed-form second-order correction quantifies that movement and gives a network-vs-control decomposition of the damping margin.

The same framework yields planning tools, but those tools are secondary to the estimator. SG-to-IBR conversion and damping or inertia placement can be ranked by post-action damping margin rather than by Fiedler participation or connectivity. Line reinforcement should be ranked by damping-margin sensitivity, not by frequency sensitivity alone. These applications require explicit comparison against existing damping-ratio placement and QEP-sensitivity methods.

The paper is therefore complete at the methodological-draft level: the derivation, reduced-model evidence, IEEE39-derived surrogate checks, ANDES baseline diagnosis, SG-to-GFL/GFM case construction, and Phase-1 calibration audit are all documented. It is not complete as a final IEEE Transactions empirical submission. The Phase-1 audit shows why: controller-only gain softening does not stabilize the low-penetration cases, while diagnostic removal of retained synchronous-machine controls stabilizes several of them. The stronger version therefore requires a defensible calibrated dynamic library, physically justified extraction of the reduced quantities, and registered comparisons against full eigensolve baselines. Until that validation is complete, the claims should be read as a mathematically grounded framework verified on surrogate QEP and reduced PLL models, not as a final empirical result for practical grids.

APPENDIX A

DERIVATION OF MODAL DAMPING SENSITIVITY

For a simple eigenpair of $\tilde{L}$, standard symmetric eigenvalue perturbation gives

$$\frac{\partial \nu_k}{\partial \rho} = q_k^T \frac{\partial \tilde{L}}{\partial \rho} q_k \quad (70)$$

and

$$\frac{\partial q_k}{\partial \rho} = \sum_{\ell \neq k} q_\ell \frac{q_\ell^T (\partial \tilde{L} / \partial \rho) q_k}{\nu_k - \nu_\ell}, \quad (71)$$

up to the usual normalization convention $q_k^T \partial q_k / \partial \rho = 0$. Since $\delta_k = q_k^T \tilde{D} q_k$,

$$\frac{\partial \delta_k}{\partial \rho} = q_k^T \frac{\partial \tilde{D}}{\partial \rho} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T (\partial \tilde{L} / \partial \rho) q_k)(q_\ell^T \tilde{D} q_k)}{\nu_k - \nu_\ell}. \quad (72)$$

Substitution into $\zeta_k = \delta_k/(2\sqrt{\nu_k})$ yields (57). The formula is ill-conditioned near repeated eigenvalues, which is precisely the case where the reduced QEP should replace the diagonal mode-by-mode sensitivity.

APPENDIX B

DERIVATION OF INERTIA SENSITIVITY

Let the inertia at bus i be the perturbation variable. Since

$$\tilde{L} = M^{-1/2} L M^{-1/2}, \quad \tilde{D} = M^{-1/2} D M^{-1/2}, \quad (73)$$

TABLE XV
FINAL CLAIM-READINESS AUDIT FOR THE CURRENT DRAFT.

Claim	Verified evidence	Control or comparison	Current status
Damping region	Exact in commuting damping; paired figure shows non-proportional screen failure.	Commuting diagonal modal equations versus coupled QEP behavior.	Theoretical/diagnostic; not a standalone novelty claim.
Second-order correction	IEEE39-derived surrogate median error improves from 0.468% to 0.032%; reduced PLL model improves median error from 2.16% to 0.09%.	Full surrogate QEP, diagonal all-mode screen, and reduced-QEP fallback.	Verified on reduced models; full ANDES IBR validation pending.
Adaptive workflow	Surrogate tails show why fallback is needed: second-order p95 9.91%, reduced-QEP p95 5.44%, adaptive p95 7.43%.	Always-diagonal, always-second-order, and reduced-QEP baselines.	Diagnostic workflow; no universal trigger theorem claimed.
SG-to-IBR conversion	Five derived ANDES IEEE39 cases load, solve power flow, and run eigenanalysis with active GFL/GFM devices; Phase-1 tested 50 case/profile controller audits.	Device inventory, power-flow/eigenanalysis success, controller-only profiles, and diagnostic removal of retained SG controls.	Case-construction and calibration audit; final estimator validation still blocked by calibrated full-IBR surrogate matching.
Weak links and reinforcement	In the IEEE39-derived surrogate, 35/46 reinforcements raise critical modal stiffness while reducing damping margin.	Frequency-only reinforcement versus finite post-action damping margin.	Surrogate planning diagnostic; not yet a full ANDES topology claim.
Inertia sign reversal	With fixed D_i/M_i , 5447/16229 perturbations reduce both the diagonal modal screen and the full-QEP damping margin.	Fixed- D_i/M_i check versus fixed- D_i direct damping-dilution control.	Reduced-model sign diagnostic; not yet a full ANDES IBR claim.
Joint inertia-damping term	Local perturbation formula shows the cross term between $E^{(D)}$ and $E^{(L)}$.	Reduces to the fixed-inertia second-order correction when $\Delta M = 0$.	Theoretical mechanism; not a finite-conversion estimator.

inertia enters both normalized operators. With $E_i = e_i e_i^T$,

$$\frac{\partial M^{-1/2}}{\partial M_i} = -\frac{1}{2} M_i^{-3/2} E_i, \quad (74)$$

and hence

$$\frac{\partial \tilde{L}}{\partial M_i} = -\frac{1}{2M_i} (E_i \tilde{L} + \tilde{L} E_i), \quad (75)$$

$$\frac{\partial \tilde{D}}{\partial M_i} = -\frac{1}{2M_i} (E_i \tilde{D} + \tilde{D} E_i), \quad (76)$$

when physical D is held fixed. For a simple eigenpair (ν_k, q_k) of $\tilde{L}$,

$$\frac{\partial \nu_k}{\partial M_i} = q_k^T \frac{\partial \tilde{L}}{\partial M_i} q_k = -\frac{\nu_k}{M_i} q_{k,i}^2. \quad (77)$$

The modal damping coefficient is $\delta_k = q_k^T \tilde{D} q_k$. Its derivative has a direct normalized-damping term and a modal-rotation term:

$$\begin{aligned} \frac{\partial \delta_k}{\partial M_i} &= q_k^T \frac{\partial \tilde{D}}{\partial M_i} q_k + 2 \sum_{\ell \neq k} \frac{(q_\ell^T (\partial \tilde{L} / \partial M_i) q_k)}{\nu_k - \nu_\ell} (q_\ell^T \tilde{D} q_k) \\ &= -\frac{q_{k,i}}{M_i} (\tilde{D} q_k)_i - \frac{1}{M_i} \sum_{\ell \neq k} \frac{(\nu_k + \nu_\ell) q_{\ell,i} q_{k,i}}{\nu_k - \nu_\ell} (q_\ell^T \tilde{D} q_k). \end{aligned} \quad (78)$$

Therefore,

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial M_i} - \frac{\delta_k}{4\nu_k^{3/2}} \frac{\partial \nu_k}{\partial M_i}. \quad (79)$$

Substitution of (77) gives

$$\frac{\partial \zeta_k}{\partial M_i} = \frac{1}{2\sqrt{\nu_k}} \frac{\partial \delta_k}{\partial M_i} + \frac{\delta_k q_{k,i}^2}{4M_i \sqrt{\nu_k}}. \quad (80)$$

The last term is the frequency-normalization effect and is positive. The first term is sign-indefinite because inertia changes both normalized damping and modal orientation. If, instead, the local damping-per-inertia ratio D_i/M_i is held fixed, the direct term $q_k^T (\partial \tilde{D} / \partial M_i) q_k$ is removed and only the modal-rotation contribution remains. This is the nontrivial sign test used in Table X.

APPENDIX C REPRODUCIBILITY CHECKLIST

Every figure and table in a final submission should record: random seed, system case, dynamic models, operating point, IBR penetration level, estimator settings, and git commit. Synthetic experiments should be separated from benchmark experiments. The final repository should include scripts that regenerate all figures from raw cases.

APPENDIX D IMPLEMENTATION DETAILS FOR SYNTHETIC TESTS

The synthetic tests in this draft can be reproduced with the following structure. Generate a connected weighted graph by first sampling a random spanning tree and then adding edges with probability p . Draw inertias M_i and damping-per-inertia ratios D_i/M_i from log-uniform distributions. Build L , compute $\tilde{L}$ and $\tilde{D}$, and compare

$$\zeta_{\text{full}} = \min_{s_j(A)} -\text{Re}(s_j)/|s_j|, \quad (81)$$

$$\zeta_{\text{diag}} = \min_k \delta_k / (2\sqrt{\nu_k}), \quad (82)$$

$$\zeta_{\text{reqp}} = \min_{s_j(P_r)} -\text{Re}(s_j)/|s_j|. \quad (83)$$

The clustered test can be produced with a nearly complete graph with almost equal edge weights. This creates repeated or nearly repeated eigenvalues and exposes the instability of mode-by-mode diagonal reasoning.

For line sensitivity, use a perturbation $\epsilon \in [10^{-5}, 10^{-3}]$ and verify that the finite-difference value stabilizes over at least one decade of ϵ . If the critical mode switches, report the switch rather than averaging incompatible derivatives. For conversion ranking, log the full candidate list, not only the winner; top- k agreement is more informative than top-1 agreement when several actions have nearly equal margins.

APPENDIX E LITERATURE SEARCH PROTOCOL

Before the word “novel” appears in a final submission, the exact claims must be searched. The search should not ask

whether damping ratios, QEPs, or line sensitivities exist; they do. The search should ask whether the specific combination exists:

- damping-region or consensus-region geometry for non-proportionally damped power-grid QEPs,
- all-mode diagonal damping screen plus coupling-triggered reduced QEP,
- damping-selected QEP reduction for IBR planning,
- SG-to-IBR conversion ranking by reduced damping margin,
- damping-aware line reinforcement using $d\zeta_{\min}/dw_e$ rather than $d\nu/dw_e$.

The minimum search set should include IEEE Xplore, Google Scholar, arXiv, and recent proceedings in power systems and control. Useful query strings include:

- “non-proportional damping power network quadratic eigenvalue damping ratio”,
- “reduced QEP damping ratio inverter power system”,
- “spectral graph damping margin inverter based resources”,
- “power system damping ratio line sensitivity eigenvalue sensitivity”,
- “consensus region damping ratio power network Laplacian”.

Each close paper should be classified into one of three categories: direct overlap, adjacent method, or background. A direct-overlap paper would force a reframe from novelty to validation/application. An adjacent-method paper should be cited and used to sharpen the distinction. A background paper supports the framing but does not reduce the contribution.

The final related-work section should include a claim table. For each claimed contribution, the table should identify the nearest prior art, the exact difference, and the evidence supplied in the manuscript. This prevents broad novelty claims and makes the contribution reviewer-checkable.

APPENDIX F

EXPECTED DATA PRODUCTS FOR SUBMISSION

A submission-ready package should include more than the PDF. It should include a reproducible data trail:

- raw case files for every network tested,
- scripts that generate L, M, D from each case,
- scripts that run the full eigensolve, diagonal estimate, and reduced QEP,
- scripts for conversion ranking and line-sensitivity ranking,
- CSV files for every figure and table,
- figure-generation scripts with fixed seeds,
- a manifest recording software versions and git commit.

For the IEEE 39-bus study, the primary tables should have the following fields. The estimator table should include penetration level, number of dynamic states, true critical pole, true $\zeta_{\min}$, diagonal estimate, reduced-QEP estimate, adaptive trigger, and relative error. The conversion table should include candidate bus, device type, post-conversion margin by full eigensolve, post-conversion margin by estimator, rank difference, and regret. The line table should include edge index,

base weight, $\partial\nu/\partial w$, analytic $\partial\zeta/\partial w$, finite-difference $\Delta\zeta$, and whether the critical pole switched.

The final paper should not hide failed cases in supplementary material. If a high-IBR case violates the estimator assumptions, it belongs in the main discussion. The practical value of the method is knowing when a cheap spectral screen is reliable and when a full model is needed.

APPENDIX G

REVIEWER CHECKLIST FOR THE FINAL VERSION

The final version should be checked against four reviewer questions before submission. First, does each numerical result identify its control? For the estimator, the controls are the homogeneous formula, the Fiedler-only formula, the diagonal all-mode formula, a frequency-selected reduced QEP, and the full eigensolve. For planning, the controls are random selection, inertia-only selection, Fiedler participation, and frequency-only sensitivity. A result without one of these controls should not be described as evidence for superiority.

Second, does each positive result identify its mechanism? A large error reduction can occur because damping is low, because the model is nearly proportional, because the critical mode is accidentally included in the reduced subspace, or because the benchmark is too easy. The manuscript should state which mechanism is active. For example, if the diagonal estimator is accurate, the correct explanation may be weak damping and separated modes, not an inherently new spectral law. If the reduced QEP improves the estimate, the explanation should be modal clustering or off-diagonal damping coupling, not generic model reduction.

Third, does each claimed contribution have an originality status? The labels should be conservative: verified and likely original after literature search, verified but known, or not yet verified. The formulas for proportional modal damping, Rayleigh line sensitivity, and QEP eigenvalue sensitivity are known tools. The candidate originality lies in their damping-region integration for IBR planning, the coupling-triggered estimator, and the damping-aware use of line sensitivity. If the literature search finds direct overlap, the paper should reframe the contribution as validation, synthesis, or application rather than novelty.

Fourth, does the paper report the negative cases? If the diagonal screen beats the reduced QEP on most benchmark cases, that is important. If damping-aware line ranking agrees with frequency ranking on IEEE 39, that is important. If the scalar surrogate fails to track the full IBR model at high penetration, that is important. These outcomes do not make the project useless; they define the validity region. A careful negative result is better than an exaggerated positive claim.

The recommended final-submission decision is therefore simple. Submit only after the IEEE 39-bus validation supplies at least one empirically supported headline claim. If C2 is strongest, title the paper around the adaptive estimator. If C4 is strongest, title it around damping-aware planning maps. If validation shows mostly boundaries and failure modes, target the paper as a rigorous methodological note or master-thesis chapter rather than a full Transactions claim.

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